

Stationarity and unit roots

Status: draft for review

Question and intuition

Why does persistence change inference? Weak stationarity requires a constant mean and autocovariances that depend on distance, not calendar time. A unit-root process accumulates innovations; shocks do not dissipate. A deterministic trend differs from a stochastic trend because detrending can remove the former without removing accumulated shocks.

Model and assumptions

$y_t = \rho y_{t-1} + u_t$. For $|\rho| < 1$, $\text{Var}(y_t) = \sigma_u^2 / (1 - \rho^2)$ in the stationary distribution. For $\rho = 1$, $\text{Var}(y_t) = t\sigma_u^2$ conditional on a fixed initial value. An $I(1)$ series becomes stationary after one difference; an $I(2)$ series requires two.

Interpretation and common errors

ADF and PP test a unit-root null; KPSS tests stationarity. Disagreement is evidence of uncertainty, not a vote to be resolved by whichever test supports a desired result. Zivot-Andrews allows a single estimated break under a specific alternative, but does not identify a policy regime causally. Near-unit roots and short samples make both rejection and non-rejection difficult to interpret. ADF lag and deterministic terms must be reported. Seasonal adjustment and changes in index bases can affect test behaviour.

Portfolio application

In this project, baseline levels preserve standard monetary-VAR practice. The approved TR rule tests levels first and changes prices to monthly inflation only on its predeclared trigger. Marginally explosive fitted roots do not automatically invalidate finite-horizon impulse calculations. They do weaken stationary-bootstrap coverage claims. The response of inflation and the cumulative response of the price level are different objects.

Exam practice

1. What is weak stationarity?

Answer. Constant finite mean/variance and covariance depending only on lag.

2. How does a random walk differ from a trend-stationary series?

Answer. Random-walk shocks have permanent level effects; trend-stationary deviations revert.

3. Can an ADF non-rejection prove $I(1)$?

Answer. No. It may reflect low power, breaks or a wrong deterministic specification.

4. Why test the first difference?

Answer. To assess whether one difference is sufficient and whether $I(2)$ remains plausible.

5. Must every monetary VAR be differenced?

Answer. No. Levels preserve long-run information; assumptions and inference limits must be explicit.

Worked reasoning

Before interpreting an estimate, name the outcome unit, shock unit, identifying restrictions, sample and uncertainty method. If any is missing, the numerical sign alone does not support a causal claim. Separate a mechanical accounting identity from a maintained economic restriction. Explain how the answer would change under a different permissible identification without searching until a preferred answer appears. A correct exam response states what can and cannot be inferred, rather than merely naming a test or estimator.

Reference and further reading

Verified primary reference: <https://doi.org/10.2307/2938337>

These notes are original explanations written for this portfolio; no thesis text or code was used.